COURSE NOTES

Numerical Methods for Engineering

Hanfeng Zhai

Department of Mechanical Engineering, Stanford University

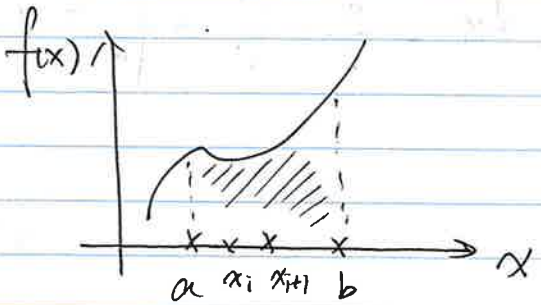
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Numerical Method Week 3.

Integration.

Rectangle Rule.



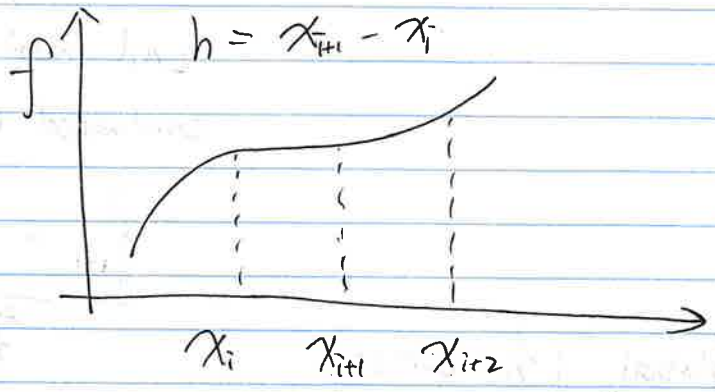
$$\int_{x_i}^{x_{i+1}} f(x)\,dx = hf(y_i) + \underbrace{\frac{h^3}{24} f''(y_i) + \cdots}_{\text{error.}}$$

$$\Big\| \quad R(f)$$

trapezoidal rule.

$$T(f) = \int_{x_i}^{x_{i+1}} f(x)\,dx = \frac{h}{2}\left[f_{i+1} + f_i\right] - \frac{h^3}{12} f''(y_i) + \cdots$$

Simpson's rule



$$\int_{x_i}^{x_{i+2}} f(x)\,dx = \frac{h}{3}\left[f(x_i) + 4f(x_{i+1}) + f(x_{i+2})\right]$$

$$\rightarrow S(f) = \frac{2}{3} R(f) + \frac{1}{3} T(f) \quad 4^{th} \text{ Order Method.}$$

(Globally)

$$= \frac{h}{2} \sum_{i=0}^{n-1} (f_i + f_{i+1}) - \frac{h^2}{12}\left(f'(b) - f'(a)\right) + O(h^4)$$

Ex

$$\int_{0.1}^{1.7} \frac{1}{\sqrt{x}}\,dx$$

9 points.

$\Delta x = h = 0.2$.

$x_i = 0.1, 0.3, 0.5, \ldots, 1.7$.

Exact.

$$2x^{1/2}\Big|_{0.1}^{1.7} = \left(2\sqrt{1.7} - \sqrt{0.1}\right) \doteq 1.97312.$$

Trapezoidal: 2.02018 ~ 2.3% (Error).

Simpson: 1.98558 ~ 0.53%

Trapezoidal w/ n correction:

1.96823 ~ 0.35%.

$\frac{1}{\sqrt{x}}$ is singular near zero, - involving derivatives brought in troubles !!!

$f(x) = e^x$. $0 \le x \le 4$. $\int_0^4 f(x)\,dx = e^4 - 1$

$= 53.59815$

"Rediscovery of the Simpson's Rule".

$$\frac{4\tilde{I}_2 - \tilde{I}_1}{3} = I + \frac{1}{4}C_2 h^4 + \frac{5}{16}C_3 h^6 + \cdots \quad (*)$$

this is a 4th order estimate of I

Continue the process.

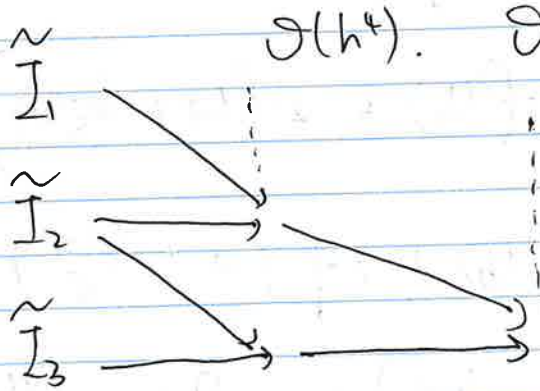
evaluate the integral with $\frac{h}{4}$ as mesh size

$$\tilde{I}_3 = I - C_1 h^2/16 - C_2 \frac{h^4}{256} - C_3 \frac{h^6}{4096} \cdots$$

$$\frac{4\tilde{I}_3 - \tilde{I}_2}{3} = I + \frac{h^4}{64} C_2 + \frac{5}{1024} h^6 C_3 + \cdots \quad (**)$$

Multiply ** by 16 & Subtract from *

$$\Rightarrow \cdots \mathcal{O}(h^6) \cdots$$



$$I_{mk} = \frac{1}{4^{m-1} - 1}$$

$$\left(4^{m-1} I_{m-k+1} - I_{m-1,k}\right)$$

$m \neq 1$

Key: With Romberg integration - you can get there exactly !!!

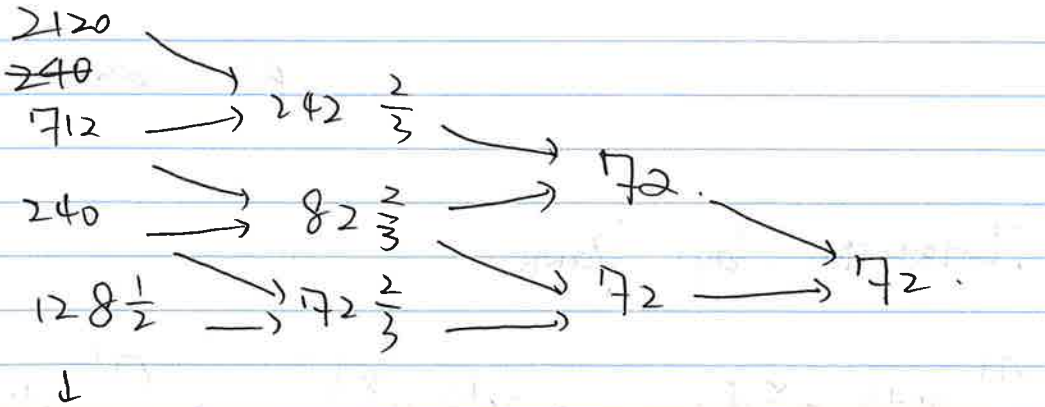
Ex. $\int_0^4 \underbrace{\left(\frac{5x^4}{8} - 4x^3 + 2x + 1\right)}_{f(x)} dx = 72$

$I_{11} = \frac{8-0}{2}\left[f(8) + f(0)\right] = 2120$

$I_{12} = \frac{2120}{2} + \frac{8}{2} f(4) = 712.$

$I_{13} = 240.$

Extrapolate: $I_{21} = \frac{1}{3}\left[4(712) - 2120\right] = 242\frac{2}{3}$



...

Brute force - you'll never get to the Exact Soln.

(Manually decreasing the mesh size).

truncation error always exist.

3) Singularity Subtraction

$$\int_0^1 x^{-1/2} e^x dx = \int_0^1 \frac{dx}{\sqrt{x}} + \int_0^1 \frac{e^x - 1}{\sqrt{x}} dx$$

$$\downarrow$$

"Can do it analytically":

$$2x^{1/2}\Big|_0^1$$

$$\lim_{x \to 0} \frac{e^x - 1}{\sqrt{x}} = \lim_{x \to 0} \frac{\sqrt{x} e^x}{2} = 0.$$

Integrals with infinite limits.

$$\int_0^\infty e^{-x^2} dx.$$

→ map or change of variables

$$t = \frac{1}{1+x} \quad \text{or} \quad t = e^{-x}$$

Get additional terms due to mapping

(adaptive quadrature — numerical, Gauss Quadrature)

Subdivide the interval (4 panels)

(formula w/ Simpson's rule)

$$S_i^{(2)} = \frac{h_i}{12}\left[f(x_i) + 4f\left(x_i + \frac{h_i}{4}\right) + 2f\left(x_i + \frac{h_i}{2}\right) + 4f\left(x_i + h_i/4\right) + f(x_i + h_i)\right]$$

Basic idea. - compare the 2 estimate.

S_i & $S_i^{(2)}$

⇓

estimate the accuracy.

If accuracy acceptable - take $S_i^{(2)}$ & go to next interval.

If not - Subdivide for next

$$I_i - S_i = c h_i^5 f^{(iv)}\left(x_i + \frac{h_i}{2}\right) + \cdots \quad (*)$$

$$I_i - S_i^{(2)} = c\left(\frac{h_i}{2}\right)^5\left[f^{(iv)}\left(x_i + \frac{h_i}{4}\right) + f^{(iv)}\left(x_i + \frac{3h_i}{4}\right)\right] + \cdots$$

Week 4.

Numerical Soln for ODE.

B.V.P. conditions @ extreme specified.

Initial Value problem. All conditions specified at same value of independent variable.



ODE: of the form.

$$F(t, y, y', y'', \dots, y^{(n)}) = 0.$$

IVP setup:

$$\frac{dy}{dt} = f(y,t). \qquad y(t=0) = y_0.$$

Goal: $y(t)$ for $0 \le t \le t_f$.

Numerical Soln: Marching.

Given a soln on $0 \le t \le t_n$, what is

$$y_{n+1} = \underbrace{y_n + h f(y_n, t_n) + \frac{h^2}{2} y_n''}$$

2nd order approx.

"Forward Euler"

$$y'' = \frac{dy'}{dt} = \frac{df}{dt} = \frac{\partial f}{\partial t}\frac{dt}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt} = f_t + f f_y$$

(chain rule; $\frac{dt}{dt} = 1$, $\frac{dy}{dt} = f$)

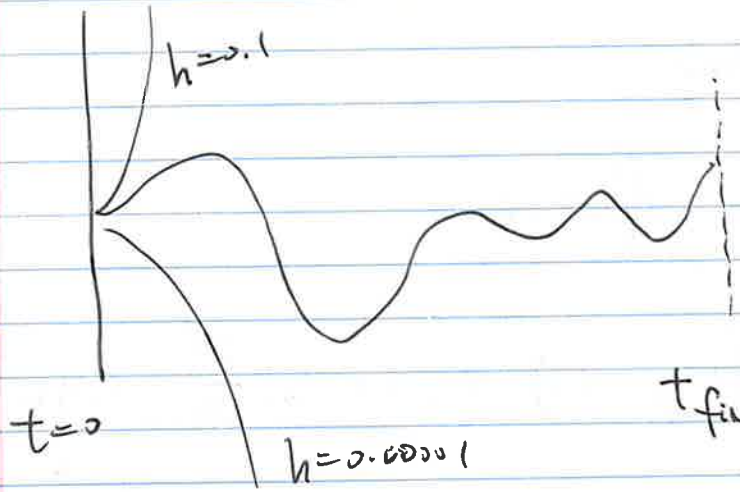
$$y''' = f_{tt} + 2 f f_{yt} + f_t f_y + f f_y^2 + f^2 f_{yy}$$

of terms increases rapidly when we consider higher order approximation.

$$f_t = \frac{\partial f}{\partial t}, \quad f_y = \frac{\partial f}{\partial y} \ldots$$

$$y' = \frac{dy}{dt} \qquad f_t = \frac{\partial^2 f}{\partial y \partial t}$$

Stability: Assuming that we are solving for a $y(t)$ that does not blow up.



↳ this numerical method is not stable

→ Stable, unstable, conditional stable.

Stable: for any choice of parameters, (Δt) does not blow up.

Unstable: for any choice of parameters, → blows up.

implications remain roughly unchanged.

Here, $y' = \lambda y$ .. the model problem.

$y' = \lambda y$. I.C.s $y(0) = 1 \Rightarrow y(t) = e^{\lambda t}$.

Consider $y'' + \omega^2 y = 0$ → Exact Solution

$$y = C_1 \cos(\omega t) + C_2 \sin(\omega t)$$

$y_1 = y$

$y_2 = y'$

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix}' = \begin{bmatrix} 0 & 1 \\ -\omega^2 & 0 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

give $\lambda^2 + \omega^2 = 0$

↳ $\lambda_\pm = \pm i\omega$

↳ eigenvalues are imaginary.

$\lambda_+ = i\omega$

$\lambda_- = -i\omega$

$$\begin{pmatrix} i\omega & 0 \\ 0 & -i\omega \end{pmatrix}$$

$A = S \Lambda S^{-1}$

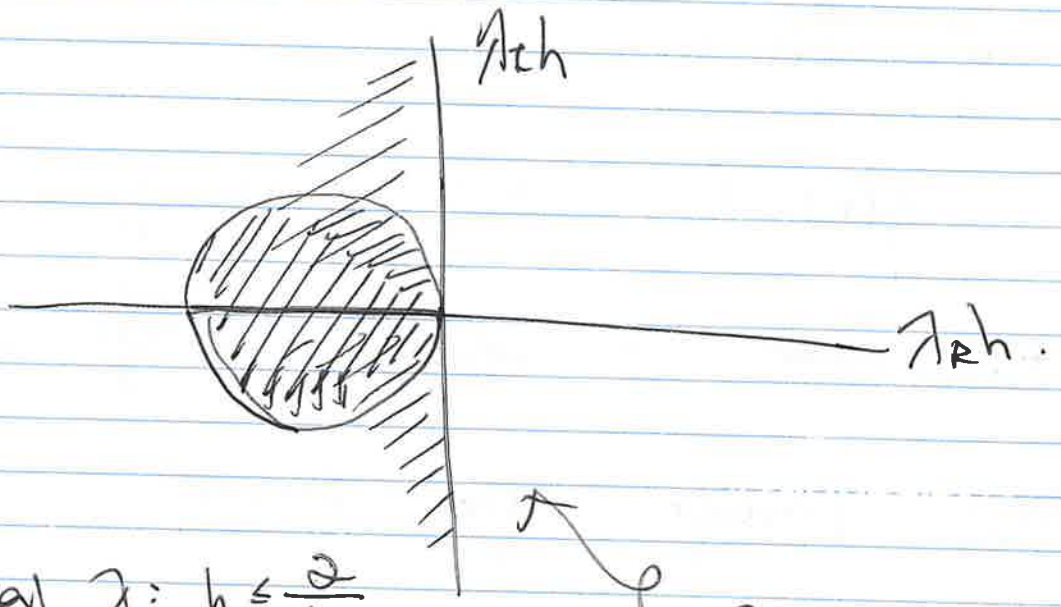
↑ eigenvalues

$\vec{y}' = S \Lambda S^{-1} \vec{y}$

$\vec{z} = S^{-1} \vec{y}$

$S^{-1} \vec{y}' = \Lambda S^{-1} \vec{y} \Rightarrow \vec{z}' = \Lambda \vec{z}$

Actual stability



Real λ: $h \leq \frac{2}{|\lambda|}$

Explicit. Euler stability region

$$y_n'' = \frac{d^2y}{dt^2} = \frac{dy'}{dt} = \frac{df}{dt} = \frac{\partial f}{\partial t} + \frac{\partial f}{\partial y}\cdot\overbrace{\frac{dy}{dt}}^{f}$$

Too many terms for high order.

Euler Method:

$$y_{n+1} = y_n + h f(y_n, t_n).$$

Locally, 2nd order.

→ Globally, 1st order.

Other Methods.

Runge - Kutta Methods.

Sol'n at t_{n+1} is obtained as a function of y_n, $f(y_n, t_n)$ & $f(y,t)$ evaluated at intermediate points.

Multi-Step Methods.

Formula involves $y_n, y_{n-1}, y_{n-2}, \ldots$
$f_{n-1}, f_{n-2}, \ldots$

Exact Soln.

$$y = y_0 e^{\lambda t}$$

$$= y_0 \left[e^{\lambda_R t + i \lambda_I t} \right]$$

$$= y_0 e^{\lambda_R t} \cdot e^{i \lambda_I t}.$$

E.g.

$$y'' + \alpha y' + \beta y = 0$$

Stability,

Euler $y_{n+1} = y_n + h f(y_n, t_n)$.

$$= y_n + \lambda h y_n + (1 + \lambda h) y_n.$$

$\sigma = 1 + \lambda h$. amplification factor.

$y_n = \sigma^n y_0$ Soln of diff eqn.

exact

$$y = y_0 e^{\lambda t}, \quad y_n = y_0 e^{\lambda n h}$$

$$|\lambda_R h| \leq 2 \Rightarrow h \leq \frac{2}{|\lambda|}$$

Stability Analysis for Backward Euler.

$$y_{n+1} = y_n + h f(y_{n+1}, t_{n+1})$$

$$y' = \lambda y.$$

$$y_{n+1} = y_n + \lambda h y_{n+1}$$

$$(1 - \lambda h) y_{n+1} = y_n \Rightarrow y_{n+1} = \underbrace{\frac{1}{1-\lambda h}}_{\sigma} y_n.$$

$$\sigma = \frac{1}{1 - \lambda_R h - i\lambda_I h}, \quad \text{For stability. } |\sigma| \leq 1$$

Quick Rev - complex numbers.

$$Z_1 = a + ib. = \sqrt{a^2+b^2}\, e^{i\theta}$$

$$\hookrightarrow \theta = \tan^{-1}\frac{b}{a}.$$

$$|e^{i\theta}| = \cos^2\theta + \sin^2\theta = 1.$$

Using model equation $y' = \lambda y$

for assessement of accuracy in addition to stability.

Exact soln: $y = y_0 e^{\lambda t} = y_0 e^{\lambda n h} = (y_0 e^{\lambda h})^n$

$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \cdots$

$= y_0 \left(1 + \lambda h + \frac{\lambda^2 h^2}{2} + \frac{\lambda^3 h^3}{6} + \cdots\right)^n$

→ Explicit Euler

$y_n = y_0 (1 + \lambda h)^n$ → convention:

compare amplification factor to the exponential expansion

⇒ α^{th} order if one recovers: up to & including $\frac{\lambda^\alpha h^\alpha}{\alpha!}$ ...

Backward Euler

$\sigma = \frac{1}{1 - \lambda h}$

$G = A e^{i\theta}$

Phase error.

$= \omega h - \theta$

$\theta = \tan^{-1} \omega h$

$= \omega h - \frac{(\omega h)^3}{3} + \frac{(\omega h)^5}{5} + \cdots$

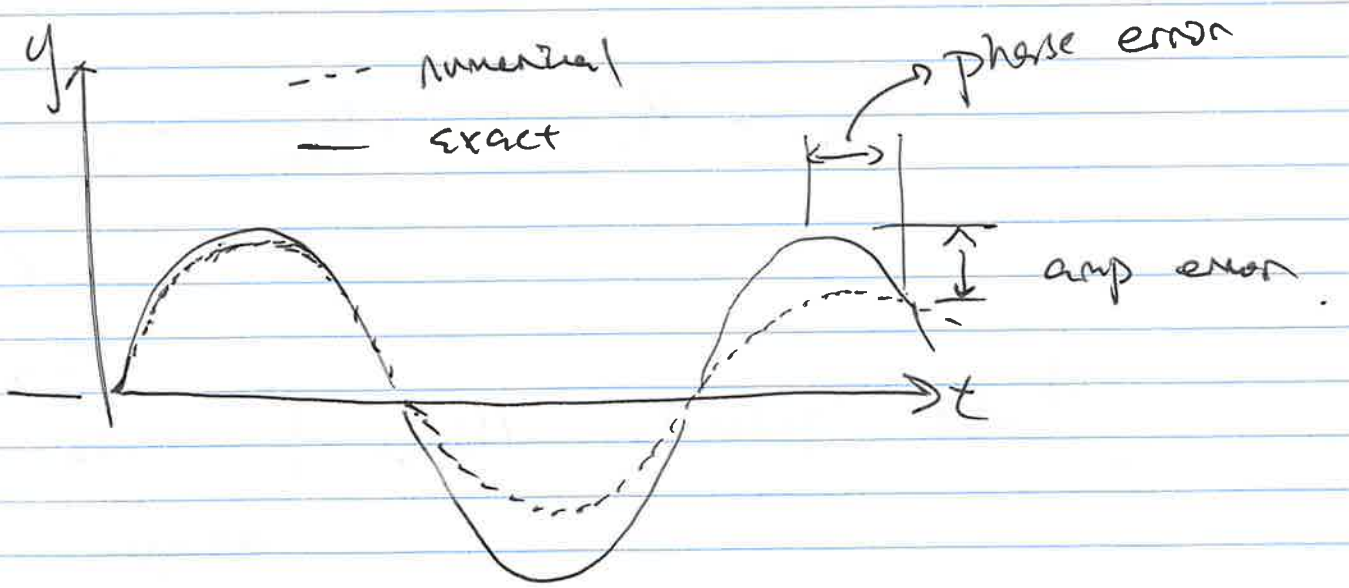
phase error.

$P.E. = \frac{(\omega h)^3}{3} + \cdots$

Exact soln:

$y = y_0 e^{i\omega t}$.

$= y_0 e^{i\omega h n}$



P.E. > 0 → phase lag.

$$\lambda = \lambda_R + i\lambda_I .$$

$$e^{\lambda h} = 1 + \lambda h + \frac{\lambda^2 h^2}{2} + \frac{\lambda^3 h^3}{3} + \cdots$$

$e^{\lambda h}$: exact soln. $1 + \lambda h$: first order accurate.

Forward & Backward Euler

Phase error

$\lambda = i\omega$. Exact s-l'n. $e^{i\omega t}$

Amp = 1

More accuracy .. trapezoidal method.

$$y_{n+1} = y_n + \frac{h}{2}\left[f(y_{n+1}, t_{n+1}) + f(y_n, t_n)\right]$$

Apply to the model equation.

$$y' = \lambda y \rightarrow y_{n+1} = y_n + \frac{h}{2}\left[\lambda y_{n+1} + \lambda y_n\right]$$

$$|\sigma| = \frac{|A|}{|B|} < 1 \qquad \text{since } |B| > |A|$$

trapezoidal Method is unconditionally stable

σ is alway ≤ 1.

Accuracy $\quad \sigma = \dfrac{1 + \lambda h/2}{1 - \lambda h/2}$

$$= \left(1 + \frac{\lambda h}{2}\right)\left[1 + \frac{\lambda h}{2} + \frac{\lambda^2 h^2}{4} + \frac{\lambda^3 h^3}{8} + \cdots\right] \qquad \frac{1}{1-x} = 1 + x + x^2 + \cdots$$

$$= 1 + \frac{\lambda h}{2} + \frac{\lambda^2 h^2}{4} + \frac{\lambda^3 h^3}{8} + \cdots + \frac{\lambda h}{2}$$

$$+ \frac{\lambda^2 h^2}{8} + \frac{\lambda^3 h^3}{16} + \cdots$$

$$= 1 + \lambda h + \frac{\lambda^2 h^2}{2} + \frac{1}{4}\lambda^3 h^3 + \cdots \qquad 2^{nd} \text{ order}$$

compare to $\quad e^{\lambda h} = 1 + \lambda h + \dfrac{\lambda^2 h^2}{2} + \dfrac{\lambda^3 h^3}{6} + \cdots$

λ real & negative.

$\lim_{n\to\infty} \sigma = -1 \Rightarrow$ soln oscillates, but does not blow up.

Backward Euler. $\sigma = \frac{1}{1-\lambda h}$.

Explicit Euler $\sigma = 1 - \lambda h$.

↖ blow up

Implicit scheme

$$y_{n+1} = y_n + h f(y_{n+1}, t_{n+1})$$

↖ @ every time step

(if nonlinear) you have to solve the algbn. eqn. (iterative - N.R., ...)

Predictor - Corrector Scheme. (RK4)

$$= y_n + \frac{h}{2}\left[\lambda(1+\lambda h) y_n + \lambda y_n\right]$$

$$= y_n \underbrace{\left(1 + \lambda h + \frac{\lambda^2 h^2}{2}\right)}$$

2nd order accuracy.

→ ... Stability, accuracy, phase error,

Runge-Kutta: 2nd order.

$$y^*_{n+1/2} = y_n + \frac{h}{2} f(y_n, t_n).$$

$$y_{n+1} = y_n + h f\left(y^*_{n+1/2}, t_{n+1}\right)$$

↖ midpoint, -inspired.

Apply to model eqn.

$y' = \lambda y$. $y^*_{n+1/2} = y_n + \frac{\lambda h}{2} y_n = \left(1 + \frac{\lambda h}{2}\right)$

Most famous algorithm of this scheme:

4th order R.K. (RK 4)

$$y^*_{n+1/2} = y_n + \frac{h}{2} f(y_n, t_n).$$

$$y^{**}_{n+1/2} = y_n + \frac{h}{2} f(y^*_{n+1/2}, t_{n+1/2})$$

$$y^{***} = y_n + h f(y^{**}_{n+1/2}, t_{n+1/2}).$$

$$y_{n+1} = y_n + \frac{1}{6} h \left[f(y_n, t_n) + 2 f(y^*_{n+1/2}, t_{n+1/2}) \right.$$

$$\left. + 2 f(y^{**}_{n+1/2}, t_{n+1/2}) + f(y^{***}_{n+1}, t_{n+1}) \right]$$

4 function evaluation @ each step.

Week 5. Lecture 2.

Runga - Kutta Method. (another notation).

$$y' = f(y,t).$$

$$k_1 = h f(y_n, t_n).$$

$$k_2 = h f(y_n + \beta k_1, t_n + \alpha h).$$

(β, α: unknown coefficients.)

$$y_{n+1} = y_n + \gamma_1 k_1 + \gamma_2 k_2.$$

(γ_1 & γ_2 are unknown)

To get the coefficients, compare to Taylor series.

$$k_2 = h\left[f(y_n, t_n) + \beta k_1 \frac{\partial f}{\partial y}\Big|_n + \alpha h \frac{\partial f}{\partial t}\Big|_n + \dots\right]$$

3 eqns for 4 unknowns.

Choose α as a free parameter

$\gamma_2 = 1/2\alpha$, $\beta = \alpha$. $\gamma_1 = 1 - \frac{1}{2\alpha}$

R-K formula:

$$K_1 = hf(y_n, t_n)$$

$$K_2 = hf(y_n + \alpha k_1, t_n + \alpha h)$$

$$y_{n+1} = y_n + \left(1 - \frac{1}{2\alpha}\right)K_1 + \frac{1}{2\alpha}K_2$$

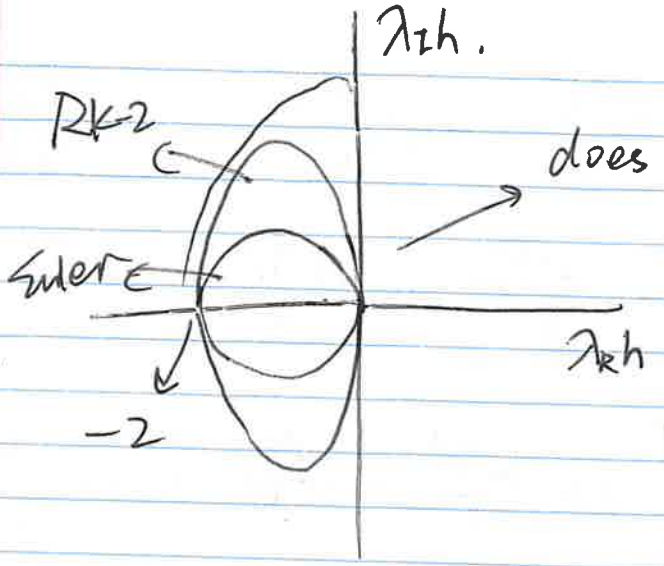
$\alpha = 1/2 \Rightarrow$ the R-K formula from last lecture

Apply this to model equation. $y' = \lambda y$.

$$K_1 = h\lambda y_n$$

$$K_2 = \lambda h[y_n + \alpha k_1] = \lambda h[y_n + \alpha h \lambda y_n]$$

$$= y_n(\lambda h + \alpha \lambda^2 h^2)$$



Euler: $\sigma = 1 + \lambda h \Rightarrow |\sigma| = \sqrt{1+\omega^2 h^2}$

Soln after 100 timesteps.

$\rightarrow y_0 |\sigma|^{100}$

ωh always < 1.
in num. meth.

* (accuracy, stability, phase error)

$\sigma = |\sigma| e^{i\theta}$.

$$\theta = \tan^{-1}\left[\frac{\text{img}(\sigma)}{\text{real}(\sigma)}\right] = \tan^{-1}\left[\frac{\omega h}{1 - \frac{\omega^2 h^2}{2}}\right]$$

$$P.E. = \omega h - \tan^{-1}\left[\frac{\omega h}{1 - \frac{\omega^2 h^2}{2}}\right]$$

$$k_3 = hf\left(y_n + \tfrac{1}{2}k_2,\ t_n + h/2\right)$$

$$k_4 = hf\left(y_n + k_3,\ t_n + h\right).$$

$$y_{n+1} = y_n + \frac{1}{6}k_1 + \frac{1}{3}(k_2 + k_3) + \frac{1}{6}k_4$$

main cost: function evaluations.

↳ Multi-Step Methods..

Formula includes information from time steps $n-1$, $n-2$, in addition to n, $n+1$ (implicit)

... Leap-frog scheme

$$y' = f(y,t)$$

$$\frac{y_{n+1} - y_{n-1}}{2h} = f(y_n, t_n)$$

Week 6 Lecture 2.

Multistep Methods.

$$y' = f(y,t), \quad y(0) = y_0$$

$t_n \to t_{n+1}$ Leap frog scheme

$$\frac{y_{n+1} - y_{n-1}}{2h} = f(y_n, t_n)$$

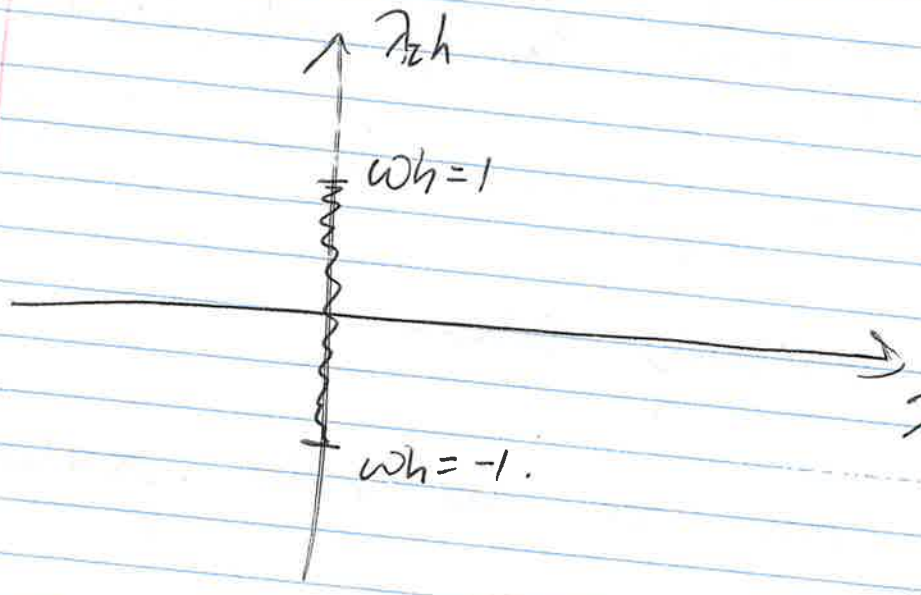
$$y_{n+1} = y_{n-1} + 2h f(y_n, t_n)$$

∴ Not self-starting.

Use another method (e.g., Euler) to go $t_0 \to t_1$

apply this to model problem $y' = \lambda y$

$$C_1 = \frac{y_1 - y_0\sigma_2}{\sigma_1 - \sigma_2} \qquad C_2 = \frac{\sigma_1 y_0 - y_1}{\sigma_1 - \sigma_2}$$



$\lambda = i\omega$

Recall: no amp. error.

2nd order Adam Bashforth.

$$y_{n+1} = y_n + h y_n' + \frac{h^2}{2} y_n'' + \frac{h^3}{6} y_n''' + \cdots$$

$$y_n' \to f(y_n, t_n) \qquad y_n'' \to \frac{h^2}{2}\left[\frac{y_n' - y_{n-1}'}{h}\right]$$

$$y_{n+1} = y_n + h f(y_n, t_n) + \frac{h}{2}\left[f(y_n, t_n) - f(y_{n-1}, t_{n-1})\right] + \mathcal{O}(h^3)$$

func eval. + cost + stability.

↑ order

Quadratic:

$$\sigma = \frac{1}{2}\left[1 + \frac{3\lambda h}{2} \pm \sqrt{1 + \lambda h + \frac{9}{4}\lambda^2 h^2}\right]$$

Recall $\sqrt{1+x} = 1 + \frac{1}{2}x - \frac{1}{8}x^2 + \frac{3}{48}x^3 + \cdots$

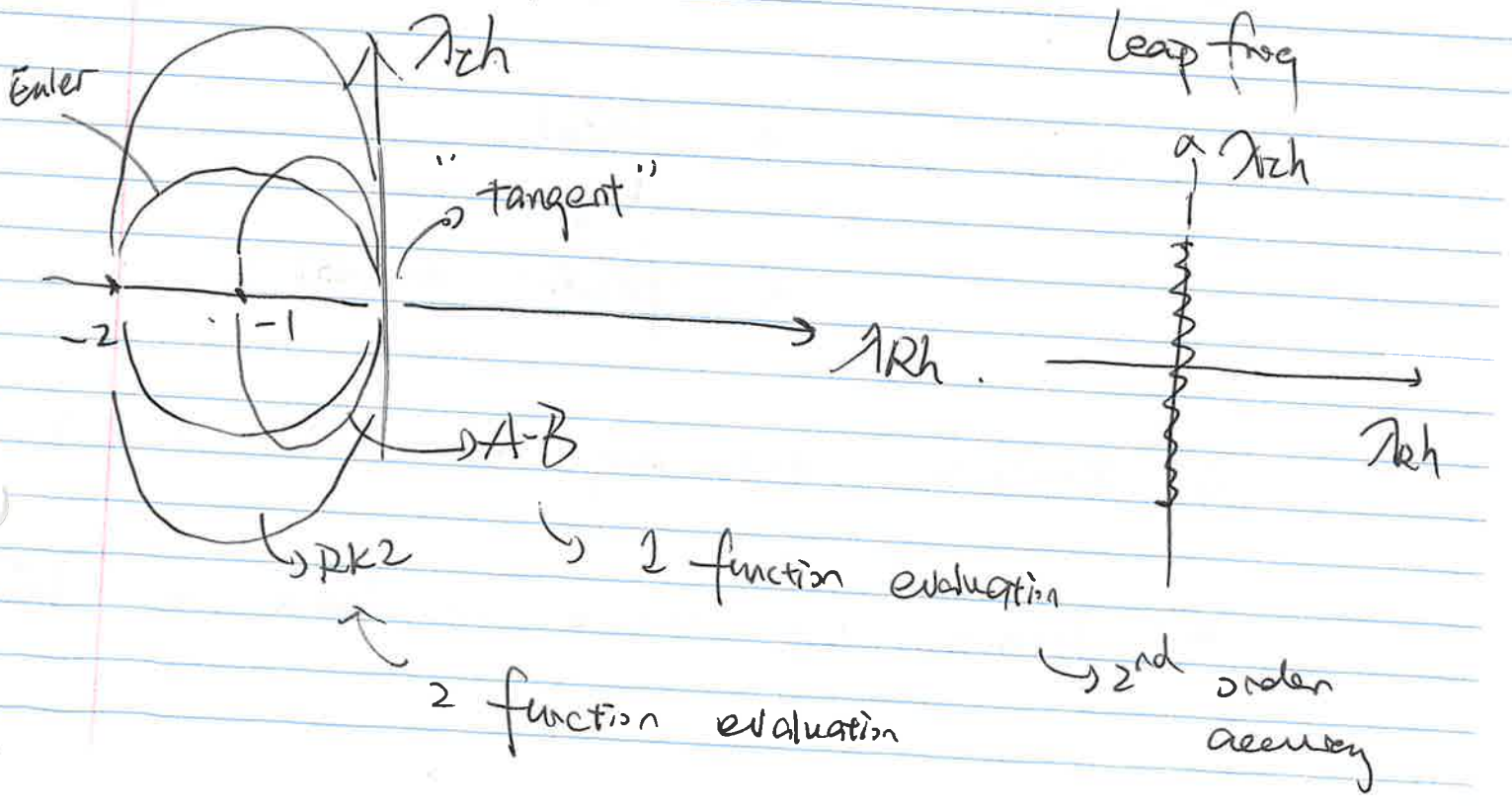
$$\sigma_1 = 1 + \lambda h + \frac{\lambda^2 h^2}{2} + \mathcal{O}(h^3)$$ real root.

$$\sigma_2 = \frac{1}{2}\lambda h - \frac{1}{2}\lambda^2 h^2 + \mathcal{O}(h^3)$$

Note that $\sigma_2 \to 0$ as $h \to 0$

... good !!!

Stability analysis.



M- evaluation numbers. #. (cost).

i.e.g., R-K4 is a 4-evaluation method per step

RK2 is a 2-eval. ...

e.g., RK-2 $h = \frac{T}{M/2}$. M/2 Steps.

Amp error $= 1 - |\sigma(k\omega h)|^{M/k}$

k: # of eval of the method per step.

Phase error.

$$\frac{M}{k}\left[k\omega h - \tan^{-1}\left(\frac{\sigma_2(k\omega h)}{\sigma_1(k\omega h)}\right)\right]$$

Leapfrog is a 1-eval method.

System of differential Equations.

^

ordinary.

High order reduce to System

$$\left.\begin{aligned} y_1 &= f \\ y_2 &= f' \\ y_3 &= f'' \end{aligned}\right\} \rightarrow \begin{cases} y_1' = y_2 \\ y_2' = y_3 \\ y_3' = y_1 y_2 \end{cases}$$

Single eqn.

$$y' = f(y,t).$$

System

$$\vec{y}' = f(\vec{y}, t)$$

$$\frac{dy_i}{dt} = f_i(y_1, y_2, \ldots, y_m, t)$$

$$i = 1, 2, 3, \ldots, m.$$

Model eqn.

$$y' = \lambda y$$

$$\frac{d\vec{y}}{dt} = A\vec{y}$$

A is $m \times m$ matrix

Going back to the model eqn.

for systems. $\vec{y}' = A\vec{y}$

$A = S\Lambda S^{-1}$ $\Longleftarrow$ constant matrix no t's.

$$\frac{d\vec{y}}{dt} = S\Lambda S^{-1}\vec{y}$$

$$S^{-1}\frac{d\vec{y}}{dt} = \Lambda S^{-1}\vec{y}$$

Let $S^{-1}\vec{y} = \vec{u}$

$$\frac{d\vec{u}}{dt} = \Lambda\vec{u} \Rightarrow \frac{du_i}{dt} = \lambda_i u_i$$

(λ_i: eigenvalues of A)

Soln: $u_i = C_i e^{\lambda_i t}$

($e^{\lambda_i t}$: modes.)

$$\vec{y} = S\vec{u} = Ce^{\lambda_1 t}\vec{S}_1 + C_2 e^{\lambda_2 t}\vec{S}_2 + \dots$$

($\vec{S}_1$: linear combinations.)

Linear case.

$$\vec{y}' = A\vec{y}$$

A: $m \times m$ matrix.

Single ODE.

$$y' = \lambda y \rightarrow \lambda = \frac{d\phi}{dy}$$

$$y' = f(y,t)$$

Euler:

$$\vec{y}^{n+1} = \vec{y}^n + hA\vec{y}^n$$

$$= (I + hA)\,y^n$$

I: identity matrix.

$$\vec{y}^n = \underbrace{(I + hA)^n}_{B}\,\vec{y}_0$$

Single eqn.

$$y^n = \underbrace{(1 + \lambda h)^n}_{\sigma}\,y_0$$

Stability. $|\sigma| \leq 1$.

$$\vec{y}^n = B^n \vec{y}_0$$

λ_i: eigs of B

For stability $|\lambda_i| \leq 1$

If need to integrate the equation, in time & time-final is large

if $\frac{|\lambda|_{max}}{|\lambda|_{min}} \gg 1 \Rightarrow$ Problem is Stiff

many time steps are required to reach t_f

e.g. System of 2 eqns.

$$u' = 998u + 1998v \qquad u(0) = 1$$

$$v' = -999u - 1999v \qquad v(0) = 1$$

$$A = \begin{bmatrix} 998 & 1998 \\ -999 & -1999 \end{bmatrix}$$

$\lambda_1 = -1$ & $\lambda_2 = -1000$

For Euler: $h_{max} = 2/1000 = \frac{1}{500}$

you can change time-steps sizes, but the round-off error will initiate the second mode → not good!

... you'll have to go with implicit method

↓

Soln to Stiffness problem

but you'll have to solve nonlinear eqn per time step

↓

Iterative methods, e.g., Newton Raphson

For non-linear $f(y,t)$. have to solve a nonlinear eqn @ each step.

⇒ inconvenience & costly.

Iterative soln methods ...

$$y_{n+1} = y_n + \Delta t\, y_n' + \frac{\Delta t^2}{2} y_n'' + \cdots$$

trapezoidal:

$$y_{n+1} - y_n = \frac{\Delta t}{2}\left[f(y_n, t_{n+1}) + \frac{\partial f}{\partial y}(y_{n+1} - y_n) + f(y_n, t_n) \right] + \vartheta(\Delta t^3)$$

$$\left[1 - \frac{\Delta t}{2}\frac{\partial f}{\partial y}\Big|_n\right](y_{n+1} - y_n) = \frac{\Delta t}{2}\left[f(y_n, t_{n+1}) + f(y_n, t_n)\right]$$

$$y_{n+1} = y_n + \frac{\frac{\Delta t}{2}\left[f(y_n, t_{n+1}) - f(y_n, t_n)\right]}{1 - \frac{\Delta t}{2}\frac{\partial f}{\partial y}\Big|_n}$$

Apply this linearization to a system of equations

Coupled set of ODEs

$$\frac{d\vec{u}}{dt} = \vec{f}(\vec{u}) = \vec{f}(u_1, u_2, \cdots, u_m)$$

↖ linearize

elements of $A_n :=$

$$\begin{bmatrix} \frac{\partial f_1}{\partial u_1} & \frac{\partial f_1}{\partial u_2} & \cdots & \frac{\partial f_1}{\partial u_n} \\ \vdots & & \ddots & \vdots \\ \frac{\partial f_m}{\partial u_1} & \frac{\partial f_m}{\partial u_2} & \cdots & \frac{\partial f_n}{\partial u_n} \end{bmatrix}$$

Apply to trapezoidal method

$$\vec{u}' = \vec{f}(\vec{u})$$

$$\frac{\vec{u}^{n+1} - \vec{u}^n}{\Delta t} = \frac{1}{2}\left[\vec{f}(\vec{u}^{n+1}) - \vec{f}(\vec{u}^n)\right]$$

$$\vec{u}^{n+1} - \vec{u}^n = \frac{\Delta t}{2}\left[\vec{f}(\vec{u}_n) + A_n(\vec{u}^{n+1} - \vec{u}^n) + \vec{f}(\vec{u}_n)\right]$$

A_n change in time

At each time step → solve linear system:

$$\left(I - \frac{\Delta t}{2} A_n\right)\vec{u}^{n+1} = \left(I - \frac{\Delta t}{2} A_n\right)\vec{u}^n + \Delta t \vec{f}(u_n)$$

solve @ every step ← linear system of algebraic eqn.

$y(0) = y_0$, & $y(L) = y_L$



Convert the 2nd order ODE → 2 1st order ODEs.

$u = y$

$v = y'$

(or in general, a system of 1st order ODE)

$$\begin{cases} u' = v \\ v' = -f(y) \end{cases} \qquad \begin{matrix} u(0) = y_0 \\ u(L) = y_L \end{matrix}$$

guess $v(0) = V_0$

Solve from $x=0$ to $x=L$ & check to see if $u(L) = y_L$ if not guess again ...

$$y_L = C_1 y_1(L) + C_2 y_2(L)$$

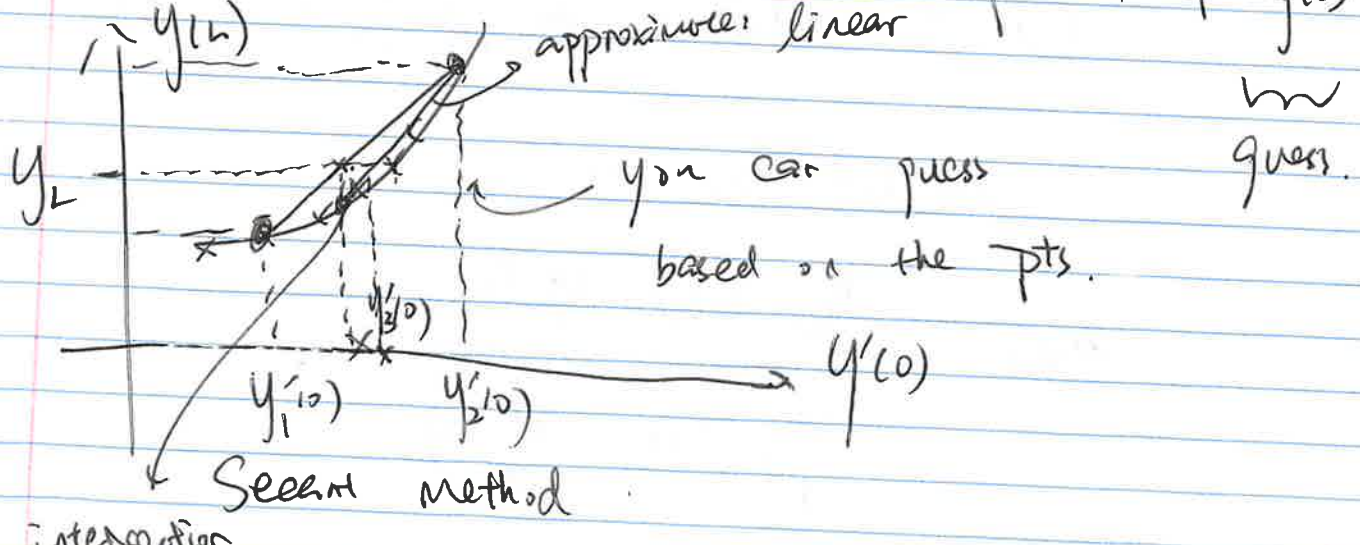
2 eqns. for C_1 & C_2

$$\begin{cases} C_1 = \dfrac{y_L - y_2(L)}{y_1(L) - y_2(L)} \\ C_2 = 1 - C_1 \end{cases}$$

Non-linear eqns. → have to iterate

e.g. secant Method

Consider $y(L)$ as a nonlinear function of $y'(0)$



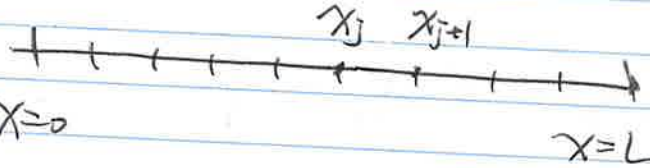
Direct Method

Approximate the derivatives in the diff. eqn. with finite differences, and solve the system of algebraic eqns.

Example. $y''(x) + A(x)\,y'(x) + B(x)\,y(x) = f(x)$

$y(0) = y_0$

$y(L) = y_L$



$h = x_{j+1} - x_j$

Second order finite differences.

$$\frac{y_{j+1} - 2y_j + y_{j-1}}{h^2} + A(x_j)\frac{y_{j+1} - y_{j-1}}{2h} + B(x_j)\,y_j = f_j$$

$N+1$ pts including the 2 boundaries

$j = 1, 2, 3, \ldots, N-1$

$$\begin{pmatrix} \beta_1 & \alpha_1 & & \cdots & 0 \\ \gamma_2 & \beta_2 & \alpha_2 & & \\ 0 & \gamma_3 & \beta_3 & \alpha_3 & \\ & & & & \\ 0 \cdots & & & \gamma_{N-1} & \beta_{N-1} \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ \vdots \\ y_{N-1} \end{pmatrix} = \begin{pmatrix} f_1 - \gamma_1 f_0 \\ f_2 \\ f_3 \\ \vdots \\ f_{N-1} - \alpha_{N-1} y_1 \end{pmatrix}$$

↑

... $3N$ operations for the tridiagonal matrix.

- Mixed boundary conditions.

$$a y_0 + b y'(0) = g$$

i) use one side of finite-diff for $y'(0)$.

$$y'(0) = \frac{-3y_0 + 4y_1 - y_2}{2h}$$

$$a y_0 + b \frac{-3y_0 + 4y_1 - y_2}{2h} = g \rightarrow$$ Solve for y_0 in terms of y_1 & y_2

P.D.E. $\longrightarrow$ semi-discretization

finite difference spatial derivatives

$\Downarrow$

System of O.D.E's.

Example

diffusion eqn. (heat eqn.)

$$\frac{\partial \varphi}{\partial t} = \alpha \frac{\partial^2 \varphi}{\partial x^2}$$

$$\varphi(0,t) = \varphi(L,t) = 0$$

$$\varphi(x,0) = g(x)$$

F-D in space



$$\frac{d\varphi_i}{dt} < \frac{\partial \varphi_i}{\partial t} = \alpha \frac{\varphi_{i+1} - 2\varphi_i + \varphi_{i-1}}{h^2}, \quad i = 1, 2, 3, \ldots, N-1$$

System of O.D.E.s

$$\frac{d\vec{\varphi}}{dt} = A\vec{\varphi}$$

A is $(N-1) \times (N-1)$ matrix

$\uparrow$

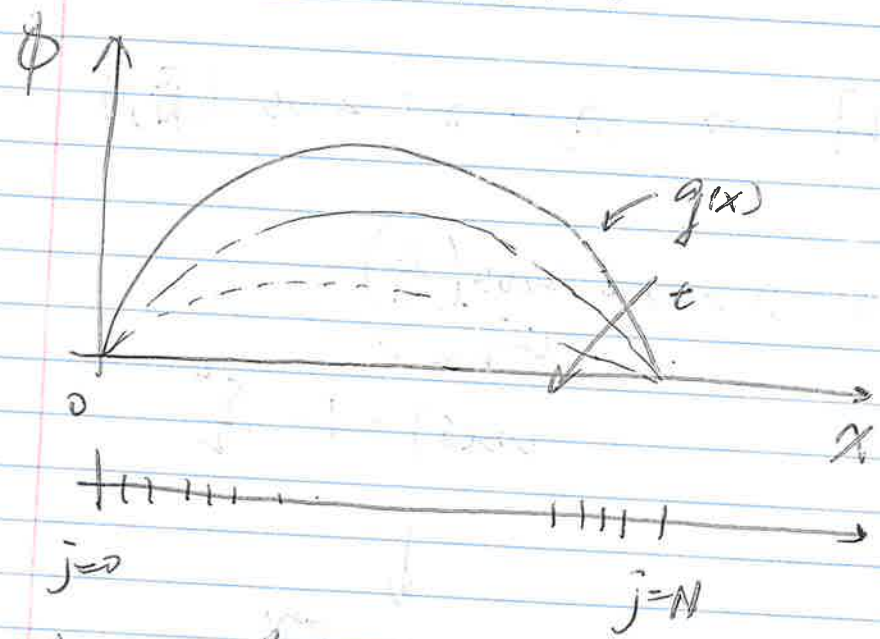
... Stiffness !

Week 8 Tue

PDEs. $\frac{\partial \phi}{\partial t} = \alpha \frac{\partial^2 \phi}{\partial x^2}$

$\phi(x,0) = g(x)$.

$\phi(0,t) = \phi(L,t) = 0$.



$$\frac{\partial \phi_j}{\partial t} = \alpha \frac{\phi_{j+1} - 2\phi_j + \phi_{j-1}}{\Delta x^2}, \quad j = 1, 2, \cdots, N-1.$$

$$\frac{d\vec{\phi}}{dt} = A\vec{\phi}, \quad \vec{\phi} = \begin{Bmatrix} \phi_1 \\ \phi_2 \\ \vdots \\ \phi_{N-1} \end{Bmatrix}$$

$$A = \frac{\alpha}{\Delta x^2} \begin{bmatrix} -2 & 1 & 0 & \cdots & 0 \\ 1 & -2 & 1 & & \\ & & & \ddots & \\ & & & & 1 \\ 0 & & & 1 & -2 \end{bmatrix}$$

$B[1, -2, 1]$

Stiffness $\frac{|\lambda|_{max}}{|\lambda|_{min}} = \frac{4\alpha/\Delta x^2}{\alpha\pi^2/\Delta x^2 N^2} = \frac{4N^2}{\pi^2}$

Stiffness increase with grid resolution.

For explicit Euler, $\lambda \in \mathbb{R}$ negative

$$\Delta t \leq \frac{2}{|\lambda|_{max}} = \frac{2}{4\alpha/\Delta x^2} \rightarrow \Delta t \leq \frac{\Delta x^2}{2\alpha}$$

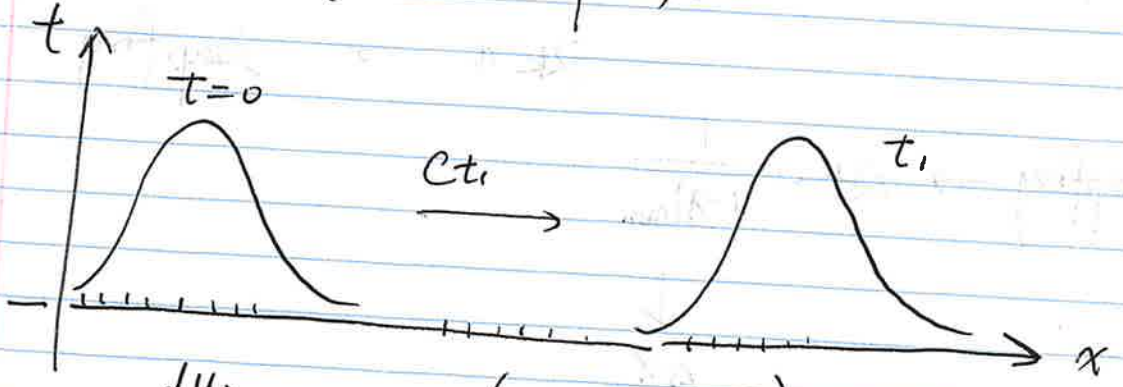
Advection Eqn.

$$\frac{\partial u}{\partial t} + c\frac{\partial u}{\partial x} = 0$$

$\downarrow$

$c > 0$

(advection speed).



$$\frac{du_j}{dt} = -c\left(\frac{u_{j+1} - u_{j-1}}{2\Delta x}\right)$$

For advection problems, stability bounds

$$\frac{C\Delta t}{\Delta x} \leq \text{const.}$$

C → CFL number

const → function of time-stepping scheme.

(Courant – Friedrich – Lewy).

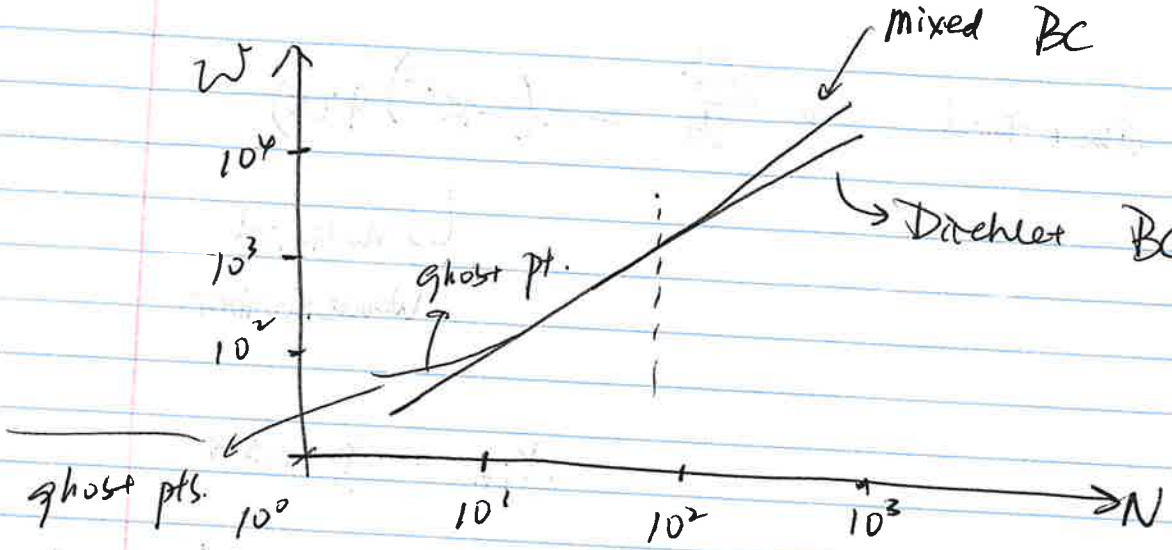
For a very slight modification to the system (e.g., B.C.s) the analysis (stability analysis) changes.

Effect of boundary conditions on stability.

$$\frac{\partial \phi}{\partial t} = \frac{\partial^2 \phi}{\partial x^2}, \qquad \phi(x,0) = g(x)$$

$$\phi(a,t) = 0,$$

$$\alpha\phi(L,t) + \frac{\partial \phi}{\partial x}(L,t) = 0$$



do not work very well for coarse mesh.

Modified wavenumber analysis (Assume P.B.C.s)

$$\frac{\partial \phi}{\partial t} = \alpha \frac{\partial^2 \phi}{\partial x^2} \qquad \text{let } \phi(x,t) = \psi(t) e^{ikx}.$$

Plug in P.D.E.

$$\frac{d\psi}{dt} e^{ikx} = \alpha(-k^2)\psi(t) e^{ikx}$$

(using the ansatz)

$$\rightarrow \frac{d\psi}{dt} = \alpha(-k^2)\psi(t).$$

↖ System decomposed !!!

at $k'\Delta x = \pi \rightarrow (k'\Delta x)^2_{max} = 4$

$$k'^2_{max} = \frac{4}{\Delta x^2}$$

Explicit Euler $\Delta t \leq \frac{2}{|\lambda|_{max}} = \frac{2}{\alpha |k'^2|_{max}} = \frac{2}{\alpha\left(\frac{4}{\Delta t}\right)}$

$$\rightarrow \Delta t \leq \frac{\Delta x^2}{2\alpha}$$

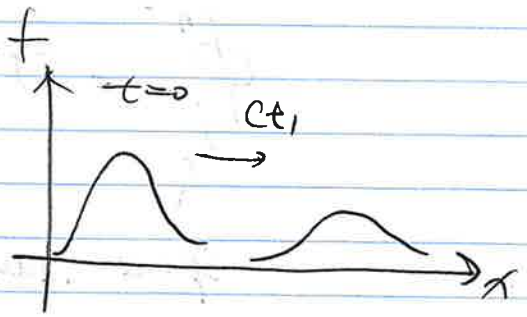
RK 4. $\Delta t \leq \frac{2.83}{\Delta t\, |\lambda|_{max}} = \frac{2}{\alpha |k'^2|_{max}} = \frac{2.79}{6}\frac{\Delta x^2}{\alpha}$.

4th-order Padé

$$(k\Delta x)^2 = \frac{12(1-\cos(k\Delta x))}{5+\cos(k'\Delta x)} = 6 \rightarrow k'^2_{max} = \frac{6}{\Delta x}$$

Advection – Diffusion

$$\frac{\partial \phi}{\partial t} = -c\frac{\partial \phi}{\partial x} + \alpha \frac{\partial^2 \phi}{\partial x^2}$$



$$\frac{d\psi}{dt} = (-ikx - \alpha k^2)\psi$$

Central 1st order

pick your scheme (eg Pdé)

Modified wave number eqn. $\frac{d\psi}{dt} = (-ic k_1' - \alpha k_2'^2)\psi$

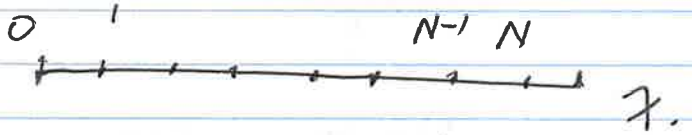
Week 8. Thurs

P.D.E.

$$\frac{\partial \varphi}{\partial t} = \alpha \frac{\partial^2 \varphi}{\partial x^2}$$

$$\frac{d\varphi_i}{dt} = \alpha \frac{\varphi_{i+1} - 2\varphi_i + \varphi_{i-1}}{\Delta x^2}$$

$i = 1, 2, 3, \ldots, N-1.$



$$\vec{\varphi} = \begin{bmatrix} \varphi_1 \\ \varphi_2 \\ \vdots \\ \varphi_{N-1} \end{bmatrix}.$$

semi-discretized.

$$\frac{d\vec{\varphi}}{dt} = A\vec{\varphi}$$

system of ODEs.

Euler time adv.

$$\vec{\varphi}^{n+1} = \vec{\varphi}^n + \Delta t A \vec{\varphi}^n$$

Similar to $y' = \lambda y$

$\hookrightarrow \lambda = -\alpha k'^2$

For 2nd order central (for 2nd derivative).

$$k'^2 = \frac{2}{\Delta x^2}\left[1 - \cos k\Delta x\right]$$

For time adv. Euler.

$$\Delta t \leq \frac{2}{\frac{2\alpha}{\Delta^2 x}\left[1 - \cos k\Delta x\right]}$$

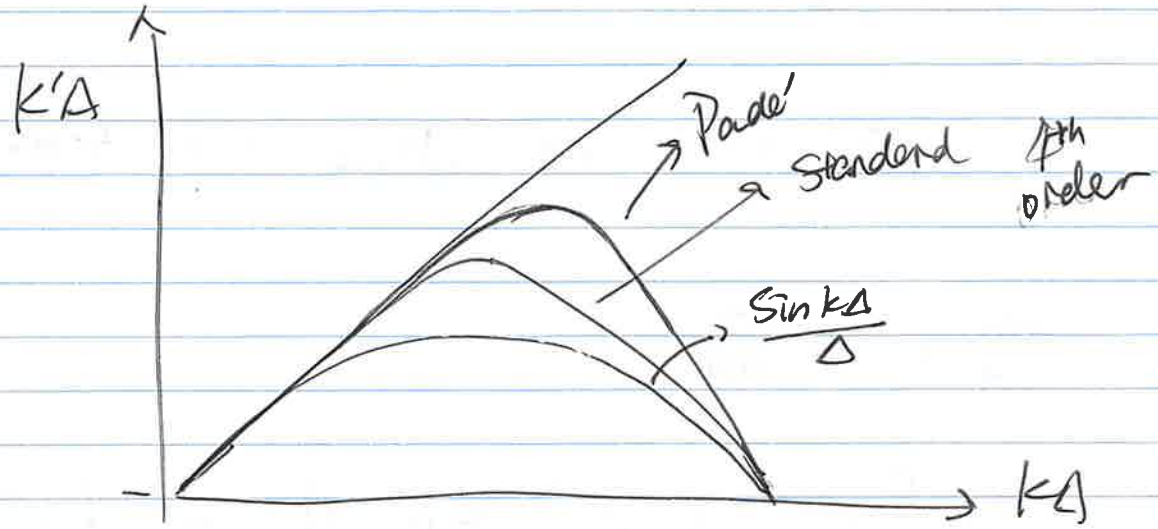
Worst case scenario. $\cos k\Delta x = -1$

$$\boxed{\Delta t \leq \frac{\Delta^2 x}{2\alpha}}$$

RK-4: $\Delta t \leq \frac{2.79\,\Delta^2 x}{4\alpha}$

Problem: $\frac{\partial u}{\partial t} + C\frac{\partial u}{\partial x} = 0$ (Convection)

For Padé's scheme → substitute new k'



Von Neumann Stability analysis

$$\frac{\partial \varphi}{\partial t} = \alpha \frac{\partial^2 \varphi}{\partial x^2}$$

Euler

$$\varphi_j^{n+1} = \varphi_j^n + \alpha \frac{\Delta t}{\Delta x^2} \left[\varphi_{j+1}^n - 2\varphi_j^n + \varphi_{j-1}^n \right]$$

Assume $\varphi_j^n = \sigma^n e^{ikx_j}$

& seek $|\sigma| \leq 1$

Heat eqn. $\frac{\partial \varphi}{\partial t} = \alpha \frac{\partial^2 \varphi}{\partial x^2}$

$$\frac{\varphi_j^{n+1} - \varphi_j^n}{\Delta t} = \frac{\alpha}{2}\left[\frac{\partial^2 \varphi^{n+1}}{\partial x^2} + \frac{\partial^2 \varphi^n}{\partial x^2}\right]$$

$$\varphi_j^{n+1} - \varphi_j^n = \frac{\alpha \Delta t}{2}\left[\frac{\varphi_{j+1}^{n+1} - 2\varphi_j^{n+1} + \varphi_{j-1}^{n+1}}{\Delta_x^2}\right.$$

$$\left. + \frac{\varphi_{j+1}^n - 2\varphi_j^n + \varphi_{j-1}^n}{\Delta^2 x}\right]$$

$$\frac{\alpha \Delta t}{2\Delta^2 x} = \beta$$

$$-\beta \varphi_{j+1}^{n+1} + (1+2\beta)\varphi_j^{n+1} - \beta \varphi_{j-1}^{n+1} = \beta \varphi_{j+1}^n + (1-2\beta)\varphi_j^n + \beta \varphi_{j-1}^n$$

$j = 1, 2, 3, \cdots, N-1$

Modified eqn. analysis.

Accuracy.

$$\frac{\partial \tilde{\varphi}}{\partial t} = \alpha \frac{\partial^2 \tilde{\varphi}}{\partial x^2}$$

$\tilde{\varphi}$ is the exact soln.

φ_j^n is the numerical soln.

(evaluated at spatial & temporal grid)

φ_a (e.g., polynomial) continuous function for the numerical soln φ_j^n.

$\varphi(x,t)$ is analytical function

↳ φ_j^n of the discrete pts.

What eqn. does φ satisfy.

Define

↳ $[\varphi]$ difference operator

If φ is soln to discrete eqn.

$$h[\varphi]$$

$$\downarrow$$

the Modified eqn.

$$\cdots - \left(\frac{\partial \varphi}{\partial t} - \alpha \frac{\partial^2 \varphi}{\partial x^2} \right) = - \alpha \frac{\Delta x^2}{12} \frac{\partial^4 \varphi}{\partial x^4} \cdots$$

which the numerical soln satisfies. is

$\tilde{\varphi}$ exact soln.

$$L[\tilde{\varphi}] \neq 0 = \underbrace{\frac{\alpha \Delta x^2}{12} \frac{\partial^4 \varphi}{\partial x^4} - \frac{\Delta t}{2} \frac{\partial^2 \varphi}{\partial x^2}}_{\text{error}}$$

$L[\varphi]$ — truncation error in the continuous case

Showed:

$$\frac{\partial \varphi}{\partial t} - \alpha \frac{\partial^2 \varphi}{\partial x^2} = \alpha \cdot \frac{\Delta x^2}{12} \frac{\partial^4 \varphi}{\partial x^4} - \frac{\Delta t}{2} \frac{\partial^2 \varphi}{\partial t^2} + \cdots$$

modified eqn.

Continous eqn. that the discrete (numerical) soln satisfies.

2nd order in space & 1st order in time

$L[\tilde{\varphi}] \neq 0$ · $\tilde{\varphi} \neq \varphi_j$ @ grid pts.

$L[\varphi] = 0$ at $\varphi = \varphi_j$ @ x_j

$$L[\tilde{\varphi}] = \underbrace{\frac{\partial \tilde{\varphi}}{\partial t} - \frac{\partial^2 \tilde{\varphi}}{\partial x^2}}_{\text{!!! } 0} - \alpha \frac{\Delta x^2}{12} \frac{\partial^4 \tilde{\varphi}}{\partial x^4} + \frac{\Delta t}{2} \frac{\partial^2 \tilde{\varphi}}{\partial t^2} + \cdots$$

(Exact Soln)

$$L[\tilde{\varphi}] = -\alpha \frac{\Delta x^2}{12} \frac{\partial^4 \tilde{\varphi}}{\partial x^4} + \frac{\Delta t}{2} \frac{\partial^2 \tilde{\varphi}}{\partial t^2} + \cdots$$

Recall $y' = \lambda y$

λ is real & negative.

$\Rightarrow$ Leapfrog will be unstable.

P.D.Es... modified wavenumber analysis.

$$\Rightarrow \frac{d\varphi}{dt} = \underbrace{-k^2}_{\lambda} \varphi$$

Ansatz: $\varphi = \varphi_{(t)} e^{ikx}$

$$\frac{\varphi_j^{n+1} - \varphi_j^{n-1}}{2\Delta t} = \alpha \frac{\varphi_{j+1}^n - 2\varphi_j^n + \varphi_{j-1}^n}{\Delta_x^2} + \mathcal{O}(\Delta_t^2) + \mathcal{O}(\Delta_x^2)$$

... unstable

D.F. $\varphi_j^n = \frac{\varphi_j^{n+1} + \varphi_j^n}{2} + \mathcal{O}(\Delta t^2)$.

resulting eqn. $\rightarrow$ unconditionally stable

Rearrange: $\beta = \alpha \frac{\Delta t}{\Delta_x^2}$

$$(1+2\beta)\varphi_j^{n+1} = (1-2\beta)\varphi_j^{n-1} + 2\beta\varphi_{j+1}^n + 2\beta\varphi_{j-1}^n$$

Stability analysis for discrete form: von Neumann

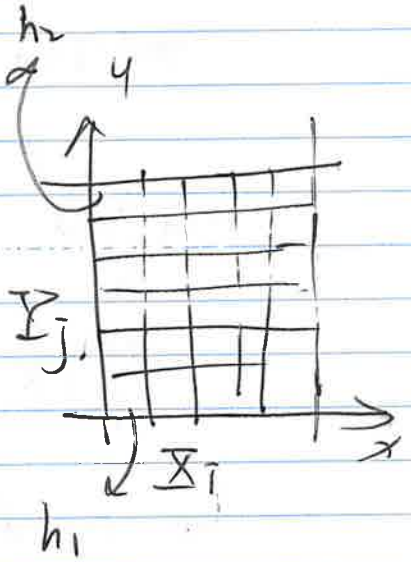
$$\frac{u^{n+1}-u^n}{\Delta t} = u^n \frac{\partial u^n}{\partial x} + \nu \frac{\partial^2 u^{n+1}}{\partial x^2}$$

Euler — backward Euler

$\dots \Delta t \le \frac{\Delta x^2}{2\nu}$.

... higher dimensions.

$$\frac{\partial \varphi}{\partial t} = \alpha\left(\frac{\partial^2 \varphi}{\partial x^2} + \frac{\partial^2 \varphi}{\partial y^2}\right)$$



Use M internal pts. (that is Y_j. not on boundaries) & N internal pts in y.

Euler in time & 2nd order F.D. in space

$$\frac{\varphi_{ij}^{n+1} - \varphi_{ij}^{n}}{\Delta t} = \alpha\left[\frac{\varphi_{i+1,j}^{n} - 2\varphi_{ij}^{n} + \varphi_{i-1,j}^{n}}{h_1^2} + \frac{\varphi_{i,j+1}^{n} - 2\varphi_{ij}^{n} + \varphi_{ij-1}^{n}}{h_2^2}\right]$$

$$\Delta t < \frac{1}{2\alpha(k_1^2 + k_2^2)}$$

$$\Delta t < \frac{2}{\alpha\left[\frac{2-2\cos k_1 h_1}{h_1^2} + \frac{2-2\cos k_2 h_2}{h_2^2}\right]}$$

Worst case scenario: $\cos k_1 h_1$ & $\cos k_2 h_2 = -1$.

gives you: $\Delta t \leq \frac{1}{2\alpha\left(\frac{1}{h_1^2} + \frac{1}{h_2^2}\right)}$

If $h_1 = h_2 = h$.

$$\Delta t < \frac{h^2}{4\alpha}.$$

In 3D: $\Delta t < \frac{h^2}{6\alpha}$.

Implicit:

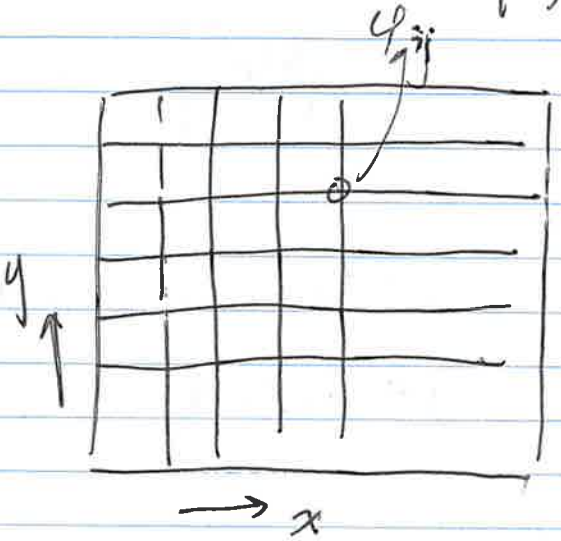
- Trapezoidal.
- Crank-Nicholson.

$$\frac{\varphi^{n+1} - \varphi^n}{\Delta t} = \frac{\alpha}{2}\left[\frac{\partial^2 \varphi^{n+1}}{\partial x^2} + \frac{\partial^2 \varphi^{n+1}}{\partial y^2} + \frac{\partial^2 \varphi^n}{\partial x^2} + \frac{\partial^2 \varphi^n}{\partial y^2}\right]$$

Week 9 Thurs.

Multi-dimensions $\frac{\partial \varphi}{\partial t} = \alpha\left(\frac{\partial^2 \varphi}{\partial x^2} + \frac{\partial^2 \varphi}{\partial y^2}\right)$.

Implicit. → have to solve a system of algebraic equations at each step.

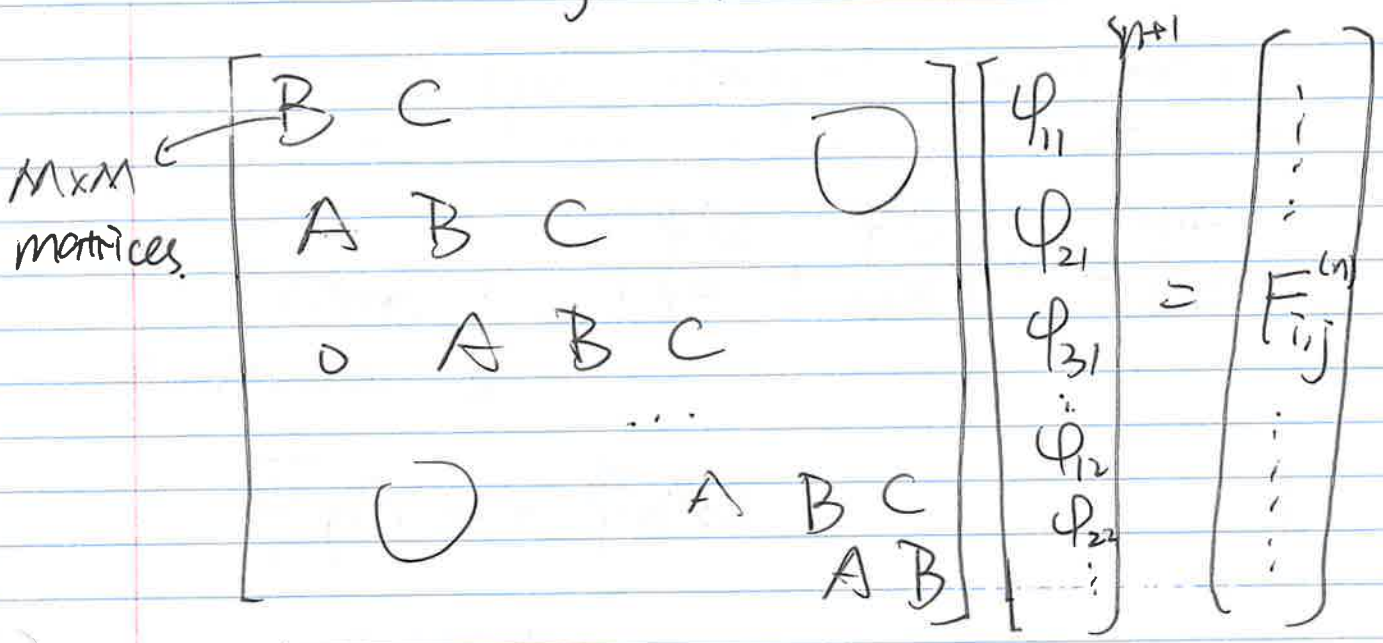


in 2D & 3D. → large matrix (sparse)

Crank-Nicholson (Trapezoidal).

$$\frac{\varphi^{n+1} - \varphi^n}{\Delta t} = \frac{\alpha}{2}\left[\frac{\varphi_{i+1,j}^{n+1} - 2\varphi_{i,j}^{n+1} + \varphi_{i-1,j}^{n+1}}{h_1^2} + \frac{\varphi_{i,j+1}^{n+1} - 2\varphi_{i,j}^{n+1} + \varphi_{i,j-1}^{n+1}}{h_2^2}\right]$$

Block – tridiagonal.



N blocks on the diagonal.

... A & C can be the same

(heat eqn.)

of arithmatic operations. $\mathcal{O}(M^2 N)$

In symbolic notation:

$$\frac{\varphi^{n+1} - \varphi^n}{\Delta t} = \frac{1}{2}\left[\frac{\partial^2 \varphi^{n+1}}{\partial x^2} + \frac{\partial^2 \varphi^{n+1}}{\partial y^2} + \frac{\partial^2 \varphi^n}{\partial x^2} + \frac{\partial^2 \varphi^n}{\partial y^2}\right]$$

$$\left[I - \frac{\alpha \Delta t}{2} A_x - \alpha \frac{\Delta t}{2} A_y \right] \vec{\varphi}^{n+1}$$

$$= \left[I + \alpha \frac{\Delta t}{2} A_x + \alpha \frac{\Delta t}{2} A_y \right] \vec{\varphi}^{n}$$

$$+ \vartheta(\Delta t) \left[\vartheta(\Delta t^2) + \vartheta(\Delta x^2) \right]$$

Approx. factorization.

$$\left(I - \frac{\alpha \Delta t}{2} A_x \right) \left(I - \frac{\alpha \Delta t}{2} A_y \right) \vec{\varphi}^{n+1} - \frac{\alpha^2 \Delta t^2}{2} A_x A_y \vec{\varphi}^{n+1}$$

$$= \left(I + \alpha \frac{\Delta t}{2} A_x \right) \left(I + \alpha \frac{\Delta t}{2} A_y \right) \vec{\varphi}^{n} - \frac{\alpha^2 \Delta t^2}{2} A_x A_y \vec{\varphi}^{n}$$

$$+ \vartheta(\Delta t) \left[\vartheta(\Delta t^2) + \vartheta(\Delta x^2) \right]$$

$$\left(I - \frac{\alpha \Delta t}{2} A_x \right) \left(I - \frac{\Delta t}{2} A_y \right) \vec{\varphi}^{n+1} =$$

$$\left(I + \frac{\alpha \Delta t}{2} A_x \right) \left(I + \frac{\alpha \Delta t}{2} A_y \right) \vec{\varphi}^{n}$$

$$+ \frac{\alpha^2 \Delta t^2}{2} A_x A_y \underbrace{\left(\vec{\varphi}^{n+1} - \vec{\varphi}^{n} \right)}_{\text{1st order T.S.}} \ldots \vartheta(\Delta t)$$

$$+ \vartheta(\Delta t) \left[\vartheta(\Delta t^2) + \vartheta(\Delta x^2) \right]$$

To get φ_{ij}, use defn of $\vec{\Sigma}$,

$$\left(I - \frac{\alpha\Delta t}{2} A_y\right) \vec{\varphi}^{n+1} = \vec{\Sigma}$$

Solve the tri-diagonal for $\vec{\varphi}$.

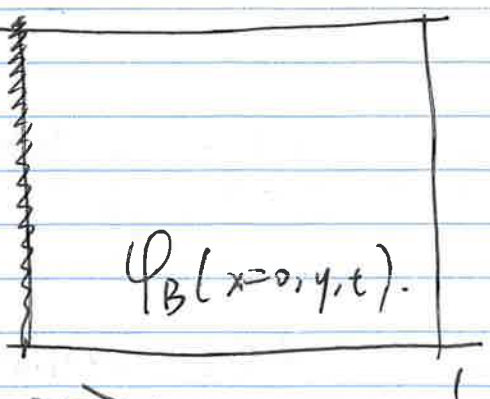
$$\varphi_{ij} - \frac{\alpha\Delta t}{2} \frac{\varphi_{i,j+1} - 2\varphi_{ij} + \varphi_{i,j-1}}{\Delta_y^2} = \Sigma_{ij}.$$

For each i, solve the tri-diagonal

the boundary conditions for Σ in terms of

B.C.s of φ...?

$\rightarrow \Sigma_{0,j}$



Recall defn of Σ. j ↑

$$\Sigma_{ij} = \varphi_{ij}^{n+1} - \frac{\alpha\Delta t}{2} \frac{\varphi_{i,j+1}^{n+1} - 2\varphi_{ij}^{n+1} - \varphi_{i,j-1}^{n+1}}{\Delta_y^2}$$

Alternating Direction Implicit Methods (A.D.I.)

$$\frac{\partial \varphi}{\partial t} = \alpha\left(\frac{\partial^2 \varphi}{\partial x^2} + \frac{\partial^2 \varphi}{\partial y^2}\right)$$

1) advance 1 step with implicit Euler acting on $\frac{\partial^2 \varphi}{\partial x^2}$ & explicit Euler on $\frac{\partial^2 \varphi}{\partial y^2}$

$$\frac{\varphi^{n+\frac{1}{2}} - \varphi^n}{\Delta t/2} = \alpha\left(\frac{\partial^2 \varphi^{n+\frac{1}{2}}}{\partial x^2} + \frac{\partial^2 \varphi^n}{\partial y^2}\right)$$

2) Advance 1/2 step with the roles of explicit & implicit scheme reversed.

$$\frac{\varphi^{n+1} - \varphi^{n+\frac{1}{2}}}{\Delta t/2} = \alpha\left(\frac{\partial^2 \varphi^{n+\frac{1}{2}}}{\partial x^2} + \frac{\partial^2 \varphi^{n+1}}{\partial y^2}\right)$$

Each step involves simple tri-diagonal (1-D solve)

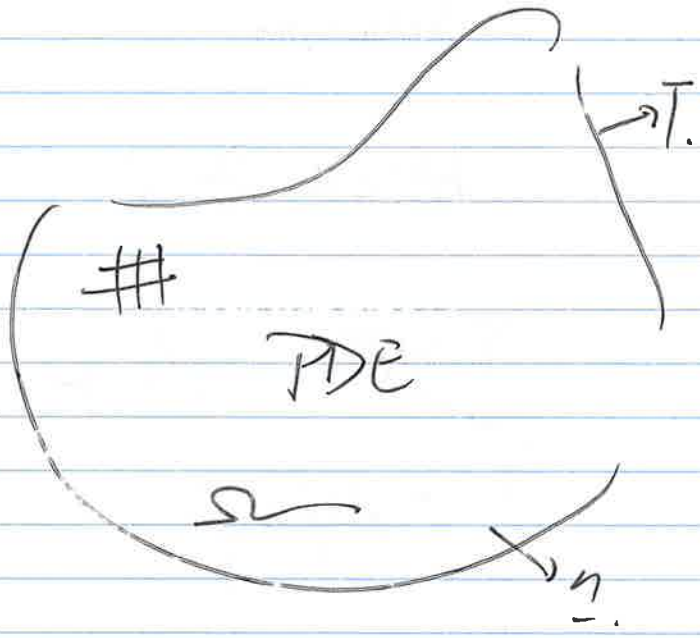
... Same as approx. factorization.

Elliptic P.D.E.

Equilibrium (steady-state).

Soln at all pts in space are coupled.

Information travels at ∞ speed.



Laplace Eqn.

$$\nabla^2 \varphi = 0. \quad \hookrightarrow \quad \frac{\partial^2 \varphi}{\partial x^2} + \frac{\partial^2 \varphi}{\partial y^2} = 0.$$

Helmholtz Eqn.

$$\nabla^2 \varphi + \alpha^2 \varphi = 0.$$

Poisson Eqn. $\nabla^2 \varphi = -f(x,y)$

Iterative Soln Method → Iterative Methods.

$$A\vec{x} = \vec{b} \quad (1)$$

$$A = A_1 - A_2$$

Iterative Scheme

$$A_1 X^{(k+1)} = A_2 X^{(k)} + b \quad (2)$$

(k → iteration index.)

Starting with initial guess $x^{(i)}$

$k = 0, 1, 2, 3, \ldots$ until convergence

Requirements.

(1) A should be easily iterable.

(2) Convergence $\lim_{k \to \infty} x^{(k)} = x$

Error $\varepsilon^{(k)} = x - x^{(k)}$

Subtract (2) from (1)

$$\Rightarrow D^{-1} = A^2 \begin{bmatrix} -1/4 & & & \\ & -1/4 & & \\ & & \ddots & \\ & & & -\frac{1}{4} \end{bmatrix}$$

$$A_2 = \begin{bmatrix} 0 & -1 & & -1 \\ -1 & 0 & -1 & \\ & \ddots & \ddots & \ddots \\ & & \ddots & \end{bmatrix}$$

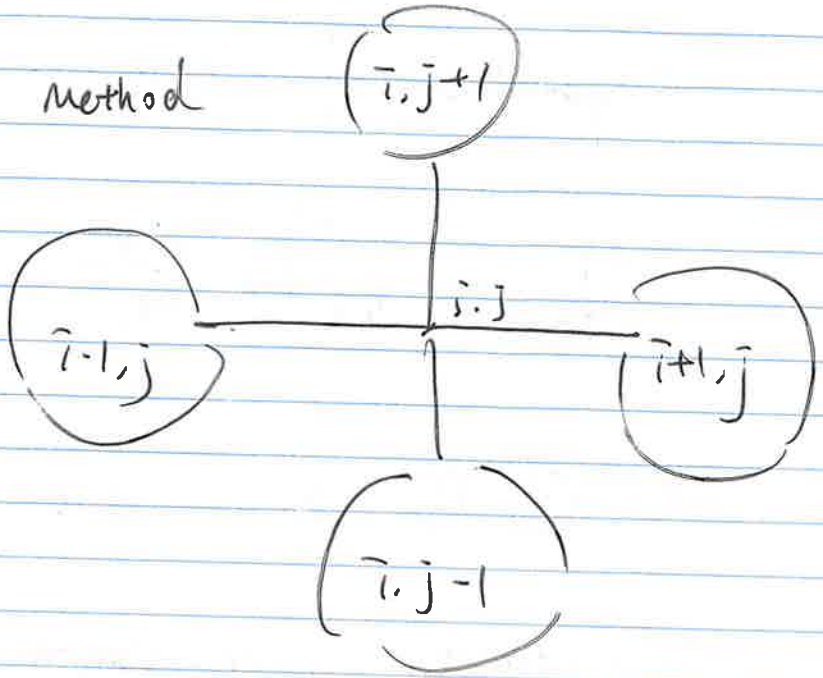
Eigenvalues of $A_1^{-1} A_2$

$$\lambda_{m,n} = \frac{1}{2}\left[\cos\frac{m\pi}{M} + \cos\frac{n\pi}{N}\right] \quad \begin{matrix} m = 1,2,3 \cdots M-1 \\ n = 1,2,3 \cdots N-1 \end{matrix}$$

$$\Rightarrow |\lambda_{m,n}|_{max} < 1$$

$\Rightarrow$ point Jacobi converges

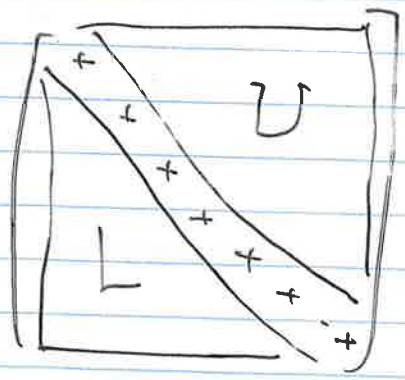
... Gauss-Seidel method



$$\varphi_{ij}^{(k+1)} = \frac{1}{4}\left[\varphi_{i-1,j}^{(k)} + \varphi_{i+1,j}^{(k+1)} + \varphi_{i,j-1}^{(k)} + \varphi_{i,j+1}^{(k+1)}\right] - \frac{\Delta^2}{4} f_{ij}.$$

In matrix notation.

$$A = \underbrace{D - L}_{A_1} - \underbrace{U}_{A_2}$$



$\omega = 1.7 \sim 1.9$

↖ fantastic improvement on convergence

Example $M = N = 30$.

	λ_{max}	Itr. #. required (factor of 1000)
Jacobi.	.9945	1270
G-S.	.989	633
SOR		46

objective: minimize λ_{max}